

Artificial Effort

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Abstract

Real-effort tasks, in which participants perform cognitively costly activities whose outcomes depend on actual performance, are widely used in experimental economics. Their validity, however, rests on the assumption that a human performs them. We study whether this assumption still holds in the era of Artificial Intelligence (AI) and Large Language Models (LLMs). Using 8 canonical real-effort tasks and 23 LLMs from three major providers, we show that most tasks can now be solved accurately and at a negligible cost, while only a few resist automation. [Benchmarking the same tasks against a group of human participants, we find that LLMs match or exceed human accuracy precisely on the tasks that are no longer resistant to automation, whereas the tasks that remain difficult for the LLMs are those on which humans substantially outperform them.](#) Within the set of models we test, mid-tier models perform close to frontier ones, broadening the set of widely accessible models that can automate these tasks. Our findings establish a boundary condition for the use of real-effort tasks in unsupervised settings: when participants can cheaply outsource task completion to an LLM, observed performance may no longer reflect genuine human effort.

Keywords: Large Language Models, Real-effort tasks, Experiment

JEL Codes: C81, C88, C90, D90.

1 Introduction

Real-effort tasks are a central methodological tool in experimental economics (Charness et al., 2018; Carpenter and Huet-Vaughn, 2019). By requiring participants to complete activities that take time, attention, and cognitive processing, these tasks are used to study effort provision, performance, fatigue, and responsiveness to incentives. Their appeal lies in the fact that outcomes are not purely stated: performance is intended to reflect the costly exertion of resources. For this reason, real-effort tasks are often treated as a relatively direct way to measure behavior-under-effort costs rather than preferences reported in the abstract. This interpretation, however, rests on an important and often implicit assumption: *the agent completing the task should be a human*.

The rapid diffusion of modern Artificial Intelligence (AI) systems and, in particular, of Large Language Models (LLMs) such as ChatGPT (OpenAI, 2022), Gemini (Pichai and Hassabis, 2023; Team et al., 2023), or Claude (Anthropic, 2023) makes this assumption less secure, especially in online environments. Indeed, in remote and unsupervised settings such as online labor markets, open survey links, and web-based experiments, researchers may be unable to verify with confidence whether a response was generated by a human participant or by an automated system (Bevendorff et al., 2025). If LLMs were able to complete canonical real-effort tasks with high accuracy at low marginal cost, then these tasks may cease to function as measures of costly human effort in precisely those contexts where they are often most attractive to use, and could also be exploited by malicious individuals as a source of revenue in a *task-milling* action. The concern is therefore not only one of participant authenticity but, rather, a question of construct validity: a task designed to capture human effort may instead capture access to a capable AI model.

This concern has become more pressing as online experimentation has expanded more rapidly than the tools available to verify participant identity and ensure proper task completion. Yet, there is still little direct evidence in the literature that state-of-the-art LLMs can effectively solve those real-effort tasks that are widely used in experimental economics, with a notable exception of how AI can replace humans in generating survey responses (Celebi et al., 2026).¹

This paper provides an assessment of whether LLMs can perform well on the canonical tasks that experimental economists routinely adopt to measure human effort. We evaluate a set of well established real-effort tasks using three families of LLMs taken from the GPT, Gemini, and Claude series and address four related research questions. First, we compare accuracy across LLMs to assess whether their performance differs and, if so, how substantially. [We also compare these results with a sample of humans that perform the task without the use of LLMs.](#) Second, we examine how performance evolves across different versions of the same model family, which is an indicator of how quickly accuracy may change over time as LLMs improve. Third, we consider the eco-

¹On the other hand, success on standard AI benchmarks—whether they test general knowledge and reasoning (e.g., MMLU (Hendrycks et al., 2021a), BIG-bench (Srivastava et al., 2022), ARC (Clark et al., 2018), HellaSwag (Zellers et al., 2019)), mathematical logic (e.g., GSM8K (Cobbe et al., 2021) MATH (Hendrycks et al., 2021b)), factual retrieval (e.g., TriviaQA (Joshi et al., 2017) and Natural Questions (Kwiatkowski et al., 2019)), or coding and agentic capabilities (e.g., HumanEval (Chen et al., 2021) and SWE-bench (Jimenez et al., 2024))—does not automatically imply success on the specific tasks economists use to elicit costly effort.

conomic dimension of substitution by comparing model performance to the cost of model use, assessing whether replacing human effort with LLM-generated output is not only feasible but also cost-effective for a profit-seeking agent.

Our results are as follows. Firstly, LLM performance varies substantially across models, determined largely by model tier rather than provider. The human benchmark reveals a second source of heterogeneity: LLMs match or outperform humans in some tasks, but humans continue to perform substantially better in others. Secondly, within each LLM tier (e.g., different versions of Claude Sonnet), newer releases always outperform previous ones with only one exception, with the largest gains concentrated in the mid-range tiers. Lastly, we show that, at typical experiment with real-effort tasks, every model costs only a small fraction of what a human participant would earn for the same workload, thereby generating a positive net gain.

2 Method

Our benchmark comprises eight real-effort tasks that are available in the Otree repository (Chen et al., 2016), implemented in their canonical form or with only minor modifications (Charness et al., 2018). Each task requires a different combination of perceptual, arithmetic, and reasoning abilities, as summarized in Table 1.²

Table 1: Summary of the eight real-effort tasks.

Task	Input	Problem
Addition (Niederle and Vesterlund, 2007)	An addition expression with 3 integer addends of up to 3 digits each.	Calculate the exact sum.
Counting Zeros (Abeler et al., 2011)	A 15×10 table of 0s and 1s.	Count the number of 0s.
Letter Decoding (Erkal et al. 2011;)	A table mapping 10 letters to the digits 0–9, together with a 7-digit coded string.	Decode the string back into the corresponding letters.
Number Pair Summation (Rosaz et al., 2016)	A 4×4 grid of decimal numbers.	Identify and return the one pair that sums to 10.0.
Sequence Completion (Bracha and Fershtman, 2013)	A sequence of numbers following a mathematical pattern.	Predict the next element.
Sudoku (Calsamiglia et al., 2013)	A partially filled 6×6 Sudoku grid (with 2×3 boxes).	Identify all missing numbers and report them in reading order, space-separated.
Distorted Text (Augenblick et al., 2015)	An image of a 12-character alphanumeric string rendered with random distortions.	Transcribe the text exactly (case-insensitive).
String Entry (Kessler and Norton, 2016)	A 9-character string of special characters (/ \) (_ < and space) displayed as an image.	Reproduce the exact string, including spaces.

²The pre-registration plan is available on OSF at <https://osf.io/7pz4m/overview>. The project website summarizing the main results is available at <https://belerico.github.io/artificial-effort/>.

2.1 Tested Models and Parameters

Our experiments evaluate 23 state-of-the-art Large Language Models (LLMs) from three major providers—OpenAI, Google, and Anthropic—spanning a range of capabilities from lightweight to frontier reasoning models. All selected models support multimodal (text and image) inputs and visual reasoning. To ensure a standardized interface, all models are accessed via the OpenRouter API.³ The full list of models, release dates, and pricing details are reported in Appendix C (Table 4).

LLMs process inputs in tokens—sub-word text units or visual patches—whose granularity varies across providers.⁴ Pricing depends on the number of input and output tokens. For reasoning-capable models (LLMs that produce an internal chain of thought before outputting their answer), intermediate reasoning tokens (i.e., the internal chain of thought, which is not shown in final the output) may also be generated and are billed at output-token rates.

To ensure comparability, we impose uniform parameters across models: we set a maximum of 2,048 output tokens (including reasoning tokens) and a time limit of 120 seconds; responses exceeding either constraint are treated as incorrect. For models with adaptive reasoning budgets, we set the reasoning effort to "high", allowing up to 80% of the tokens within the limit to be allocated to reasoning. We set the sampling temperature (a parameter which governs the concentration of the probability distribution used for token selection) to 0.6 for models that expose it (Anthropic’s and Google’s).⁵ We do not set top- p (a parameter that limits sampling to the most probable candidate tokens), which OpenRouter defaults to 1.0, and we do not set a random seed. Each model API call is independent and stateless, carrying no history from previous calls. With non-zero temperature, the outputs of an LLM call are therefore stochastic rather than deterministic. However, because LLM providers may update the model snapshot associated with a given identifier at any time, strict distributional equivalence across calls made at different points in time cannot be guaranteed.

2.2 Experimental Setup

In what follows, we consider the eight real-effort *tasks* listed in Table 1, namely: *Addition*, *Pair Summation*, *Letter Decoding*, *Sequence Completion*, *Distorted Text*, *String Entry*, *Counting Zeros*, *Sudoku*. A *trial* is one instance of a task, completed once by one model. For every model–task pair, we perform 20 trials.

Our pipeline works as follows. For each task, the oTree back-end generates 20 trials, each rendered as a standard HTML page. We grab the rendered input (textual instructions and the image representing the task instance) and send it to the LLM through the OpenRouter API. The LLM’s textual response is parsed and passed back to the oTree back-end for correctness verification.⁶ An answer is scored as correct only

³<https://openrouter.ai>

⁴<https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-count-them> and <https://ai.google.dev/gemini-api/docs/tokens>

⁵OpenAI reasoning models do not support a temperature control, so the parameter is omitted and OpenRouter applies a default of 1.0.

⁶This workflow mirrors how a human participant could use an LLM in practice: screenshotting the task page and copying the LLM’s answer into the oTree input field. Fully automating this interaction—e.g., via browser-automation tools such as Playwright (<https://playwright.dev>) is feasible and would

if it exactly matches the ground-truth solution; no partial credit is awarded. From the fraction of correct answers, we compute the accuracy of each model to solve each task. We also collect the total number of tokens used to elaborate the answer, as well as quantify the cost of performing each task.

To establish how difficult these tasks are for humans—and thus whether LLMs threaten their validity—we administered the same eight tasks to a sample of university students in a lab experiment across two different days. Participants completed all tasks through the standard oTree interface, with 10 trials per task. To incentivize performance, participants received a €5 show-up fee for each session attended. In addition, for each session, one task was randomly selected for payment, and participants earned €0.50 for each correct trial within the selected task. Human responses were scored with the same exact-match criterion applied to the tested models, and each participant’s per-task accuracy was computed as the fraction of correct trials. Given the length of the experiment, participants completed five tasks on the first day and were then invited to a second session two days later to complete the remaining three tasks. On Day 1, 36 participants took part in the experiment. Of these 36 participants, 33 also attended Day 2, along with 6 additional participants. In total, we collected 297 student-task observations.

3 Results

This section proceeds in three steps: which tasks LLMs find easy or hard relative to humans (Section 3.1), how performance varies across providers and tiers (Section 3.2), and the cost–accuracy trade-off that governs substitution (Section 3.3). Except in Subsection 3.1, where we compare LLM and human performance on the tasks, all reported performance figures refer to LLMs.

3.1 Result 1: Task difficulty

We examine how LLM accuracy varies across real-effort tasks and, unlike the subsequent analyses, benchmarks model performance against human participants. This comparison is useful because low LLM accuracy alone does not necessarily imply that a task is intrinsically difficult: a task may be easy for humans but hard for current models because it exploits weaknesses in visual parsing, enumeration, exact transcription, or multi-step reasoning. Figure 1 compares average LLM accuracy across the 23 models with human accuracy for each real-effort task. The results show substantial heterogeneity in both absolute LLM performance and the LLM–human gap. The p -values shown in Figure 1 refer to tests of the null hypothesis that the LLM–human gap is zero within each task; the underlying regression-based analysis is reported in Appendix A.

LLMs perform at or near ceiling on tasks involving short, well-structured inputs: *Addition* (100%), *Number Pair Summation* (97.6%), *Letter Decoding* (94.1%), and *Sequence Completion* (90.0%). These tasks can be solved through a limited sequence of elementary operations, such as adding small numbers, searching over candidate pairs,

add negligible latency and no additional token cost. For an alternative, albeit technically heavier, implementation via LLM-driven agents, see Celebi et al. (2026). See Appendix B for a reference implementation.

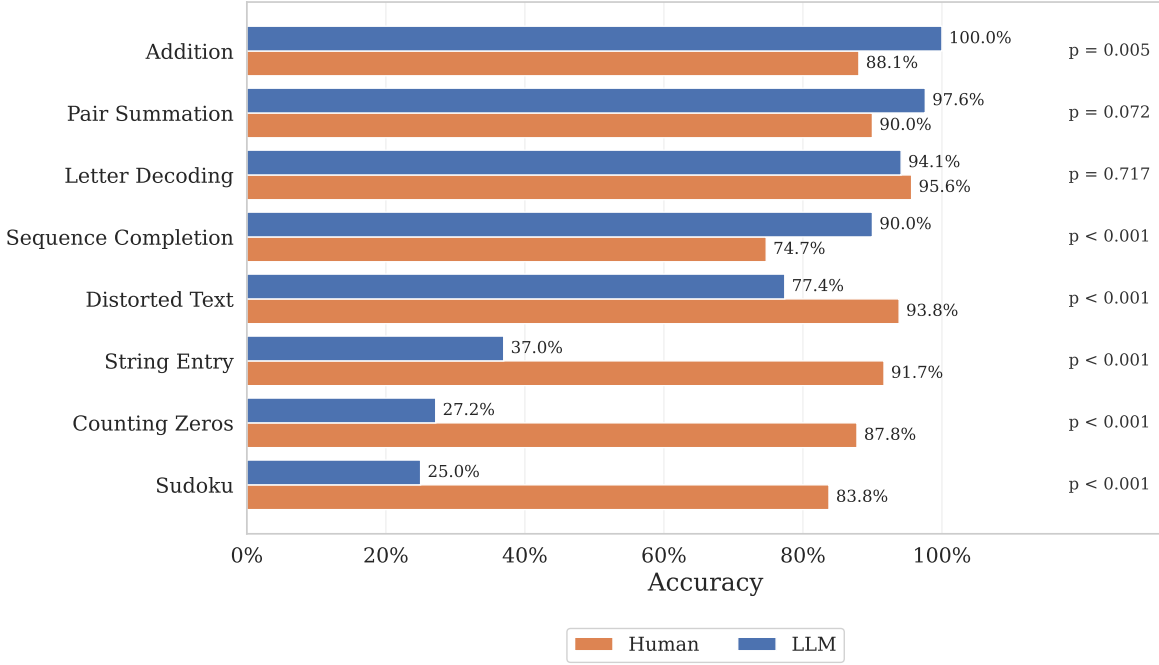


Figure 1: Average accuracy by task. Bars report mean task accuracy for humans and LLMs; p -values test the null hypothesis that the LLM–human gap is zero.

applying a lookup table, or recognizing a short pattern. Consistent with this, LLMs significantly outperform human participants on *Addition* and *Sequence Completion*, by 11.9 and 15.3 percentage points respectively. Their advantage on *Number Pair Summation* is smaller and only marginally significant (7.6 percentage points, $p = 0.072$), while performance on *Letter Decoding* is statistically indistinguishable from humans (−1.5 percentage points, $p = 0.717$). These results suggest that tasks traditionally viewed as cognitively demanding in the behavioral economics literature are, in some cases, already highly vulnerable to LLM automation (Nogueira et al., 2021; Wei et al., 2022).

By contrast, LLMs perform poorly on *Sudoku* (25.0%), *Counting Zeros* (27.2%), and *String Entry* (37.0%), all of which humans solve with high accuracy. The corresponding human–LLM gaps are large and statistically significant: 59 percentage points for *Sudoku*, 60.6 percentage points for *Counting Zeros*, and 55 percentage points for *String Entry* ($p < 0.001$ in all cases). These failures reflect different limitations. *Sudoku* requires multi-step reasoning across rows, columns, and boxes, making it a combinatorial problem that remains difficult for LLMs regardless of input format (Maiya et al., 2025). *Counting Zeros* requires reliable enumeration over a large grid, a known weakness of current architectures (Yehudai et al., 2024; Fu et al., 2024). *String Entry* requires exact reproduction of special characters and spaces, which LLMs often silently normalize or reformat, producing near-matches that score poorly.

Finally, *Distorted Text Transcription* falls between these two groups. LLM accuracy is relatively high (77.4%), consistent with the fact that character recognition is largely tractable for neural networks (LeCun et al., 2002), but still significantly below human accuracy by 16.5 percentage points ($p < 0.001$). This gap likely reflects the perceptual noise introduced by random distortions.⁷

⁷Our results align with recent evidence that multimodal LLMs handle text and object-recognition

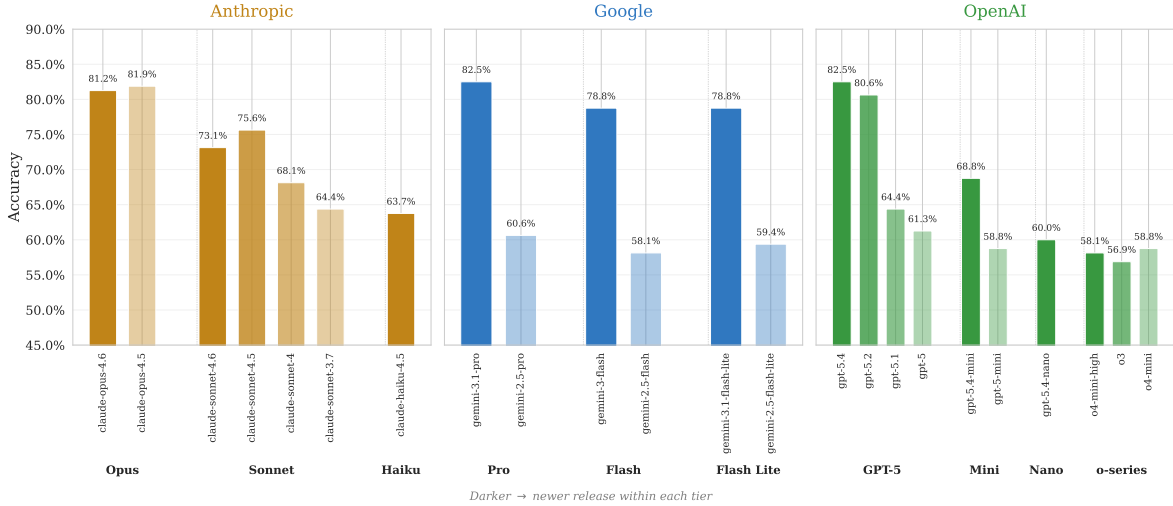


Figure 2: Average accuracy by LLM within each provider family, ordered chronologically by release date within each models tier. Each bar represents a single model’s accuracy averaged over all tasks.

Overall, these patterns show that low LLM performance does not simply identify tasks that are intrinsically difficult for humans. Humans remain highly accurate precisely on several tasks where LLMs fail. This distinction is methodologically important: some cognitively demanding tasks are now easy to automate, while some visually or procedurally simple tasks continue to function as valid measures of human effort because they exploit weaknesses of current LLMs.

3.2 Result 2: Performance across providers, tiers, and tasks

Figure 2 reports the average accuracy across the eight real-effort tasks for all 23 LLMs we tested, with models grouped by provider and tier; darker shades indicate newer releases within the same tier. Figure 3 reports the accuracy for every model–task pair. From the figures, two main patterns emerge.

First, while the best top-tier models—Claude Opus 4.5 and 4.6, Gemini 3.1 Pro, and GPT-5.4—achieve average accuracies above 80%, mid-tier ones (Claude Sonnet, Gemini Flash, GPT Mini) are not as strong but still achieve a good performance: the difference, per provider, between a best-performing mid-tier LLM and a best-performing top-tier one varies between 4% and 14%. This means that the set of models that perform well on real-effort tasks is expanding from the more-expensive top-tier models to cheaper and more widely accessible ones. *This suggests that, as providers continue to release stronger models, tasks that are currently resistant to automation may not remain so in the near future. For instance, Gemini Pro improved by 22 percentage points between versions 2.5 and 3.1, GPT-5 by 21 percentage points from GPT-5 to GPT-5.4, and Claude Sonnet by 11 percentage points from version 3.7 to 4.5 (though version 4.6 shows a slight regression).*

Second, as shown in Figure 3 and reflecting the analysis in Section 3.1, an LLM

CAPTCHAs reasonably well, but struggle with those requiring image-based, spatial, or multi-step reasoning (Ding et al., 2025; Wang et al., 2025; Zhang et al., 2025).

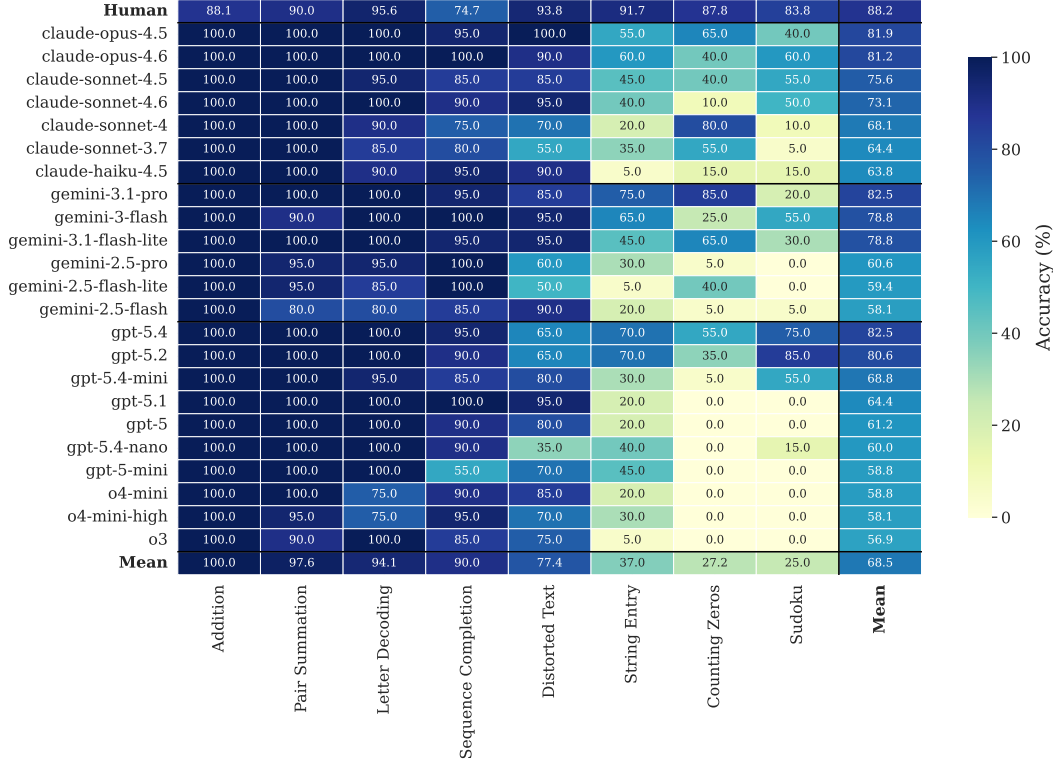


Figure 3: Task accuracy (%) per LLM. LLMs are grouped by provider family (Anthropic, Google, OpenAI) and ordered by accuracy within each family (best at the top); tasks are ordered by difficulty (easiest on the left). Black lines separate provider families. The bottom *Mean* row averages over models, and the *Human* row reports the average accuracy of human participants on each task, computed over everyone who attempted that task (36–40 participants, depending on the task).

averaging 80% overall may score near-perfectly on some tasks (e.g., *Addition* and *Pair Summation*), while failing on others (e.g., *Sudoku* and *Counting Zeros*). This indicates that some tasks are currently resistant to automation, but also suggests that the set of tasks that are not will expand with new model releases.

3.3 Result 3: Cost–accuracy trade-offs and the economics of substitution

Beyond accuracy, the practical viability of LLM-based automation depends on the cost of running the LLMs incurred by the profit-seeking agent. Figure 4 visualizes, for each model, the trade-off between the average per-task cost and the per-task accuracy. We define the average per-task cost for model m as

$$\bar{C}_m := \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} C_{m,t} \quad C_{m,t} := \sum_{i=1}^N \left(p_m^{\text{in}} \cdot T_{m,t,i}^{\text{in}} + p_m^{\text{out}} \cdot T_{m,t,i}^{\text{out}} \right),$$

where N is the number of trials, $T_{m,t,i}^{\text{in}}$ and $T_{m,t,i}^{\text{out}}$ are the input (textual prompt plus the image representing the particular task instance) and output (reasoning plus answer)

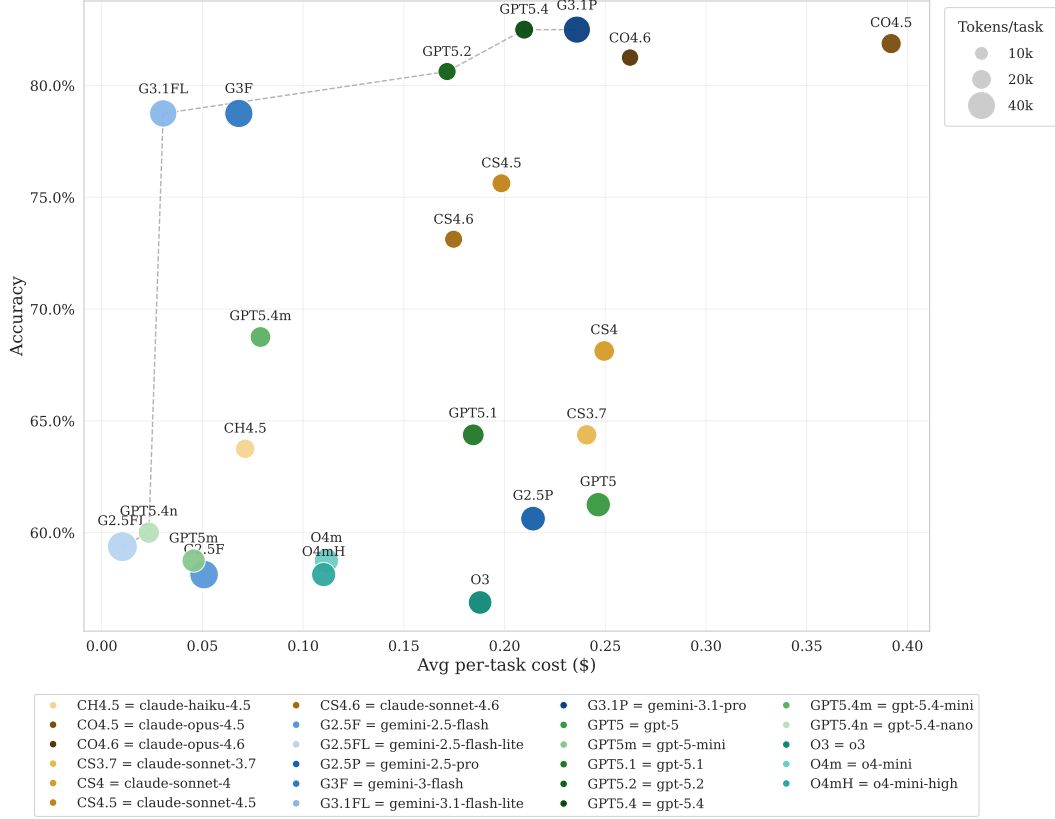


Figure 4: Cost–accuracy–tokens trade-off. Each bubble represents one model. All three quantities are per-task averages: for each model, the per-task value (computed over the 20 trials of one task) is averaged across the 8 tasks. Horizontal axis: average per-task API cost $\bar{C}_m$ (in dollars). Vertical axis: average per-task accuracy (proportion of correct answers). Bubble area: average per-task tokens (prompt plus output).

tokens for model m on the i -th trial of task t , p_m^{in} , p_m^{out} are the per-token input and output prices for model m , and $\mathcal{T}$ is the set of tasks.

The cost to achieve the (current) best performance, as shown in Figure 4, is modest in absolute terms: GPT-5.4 and Gemini 3.1 Pro achieve the highest per-task accuracy (82.5%) at an average per-task cost $\bar{C}_m$ less than \$0.25—a fraction of what a human participant would earn on a typical experiment in an platforms like Prolific or MTurk for an equivalent workload. At the same time, the cheapest LLMs still achieve an accuracy of 58–60%, meaning that even the most budget-constrained automation attempt would produce correct answers on the majority of tasks. Notably, Gemini 3.1 Flash Lite Preview achieves 78% accuracy at an average per-task cost $\bar{C}_m$ of \$0.03, suggesting that the cost barrier to high-quality automation is already negligible.

3.3.1 Optimizing performance

Notably, the assumption, made in Figure 3, of a single LLM being used for all tasks may lead to an *underestimation* of the threat of automation. This is because a strategic participant could select different LLMs for different tasks—using, for example, Gemini 3.1 Pro for *Counting Zeros* (85%), GPT-5.2 for *Sudoku* (85%), and Gemini 3.1 Pro

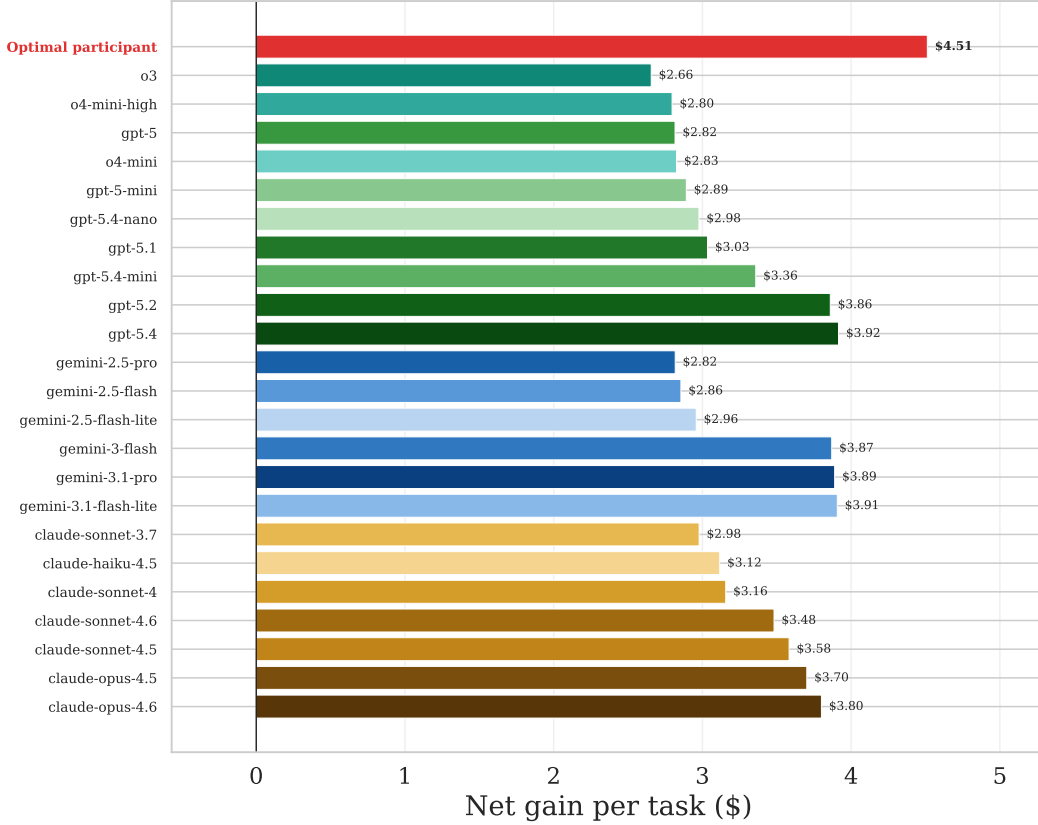


Figure 5: Average net gain $\bar{G}_m$ per-task for each model m : for each model–task pair, \$0.25 times the number of correct answers over its 20 trials minus the API cost of those trials, averaged across the 8 tasks. LLMs are grouped by provider family and ordered by net gain within each family. The red bar shows the “optimal participant” who cherry-picks the best model for each task. Every model generates positive net value.

for *String Entry* (75%). By cherry-picking the best-performing model for each task, the participant would achieve an average accuracy of 93.1%—an 11% improvement over the choice of a single model (Gemini 3.1 Pro and GPT-5.4, 82.5%). Under this “optimal participant” scenario, even the hardest tasks in our benchmark (*Sudoku*, *Counting Zeros*, and *String Entry*), which we previously identified as resistant to individual LLMs, are solved at 75% accuracy or above. Therefore, the automation resistance of a task strongly depends on the will of a knowledgeable profit-seeking participant to use the right LLM to solve it. Figure 5 formalizes this point by computing the *average per-task net gain* $\bar{G}_m$ for each model, defined as the hypothetical payout minus the LLM cost, averaged over all tasks:

$$\bar{G}_m := \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (G_{m,t} - C_{m,t}) \quad G_{m,t} := \rho \sum_{i=1}^N \mathbb{1}[\hat{y}_{m,t,i} = y_{t,i}],$$

where ρ is the payment per correct trial (we apply the rate $\rho = \$0.25$, a conservative estimate of typical online-experiment rates), $\hat{y}_{m,t,i}$ is the response of model m , and $y_{t,i}$ is the correct answer on the i -th trial of task t . Crucially, *every* model we tested generates a positive $\bar{G}_m$, ranging from \$2.66 (o3) to \$3.92 (GPT-5.4). Such a universally positive

net gain confirms that, at compensation rates typical of online experiments, replacing human participants with LLMs is not only feasible but economically dominant.⁸

4 Discussion and Conclusions

We assessed the ability of different tiers of Large Language Models from three major providers (OpenAI, Google, and Anthropic) to solve well established real-effort tasks in experimental economics. A total of 23 different models on 8 real-effort tasks were tested, along with an in-depth analysis aimed to answer four research questions. Our findings are as follows.

First, LLM performance varies substantially across models but is largely determined by model tier rather than provider. Top-tier models (Claude Opus 4.5 and 4.6, Gemini 3.1 Pro, and GPT-5.4) exceed 80% average accuracy, while less-performing models achieve around 60%. Half of the tasks in our testbed—*Addition*, *Pair Summation*, *Letter Decoding*, and *Sequence Completion*—are solved near-perfectly by all LLMs, while *Sudoku*, *Counting Zeros*, and *String Entry* remain challenging even for the best ones. The human benchmark confirms the practical reading of this split: on the near-perfectly solved tasks LLMs reach or exceed human accuracy, so these tasks are at risk of automation if administered in an unsupervised environment, whereas humans retain a large accuracy advantage on *Sudoku*, *Counting Zeros*, and *String Entry*, which therefore remain valid measures of human effort for now.

Second, within each tier, newer model releases generally improve over their predecessors: Gemini Pro improved by 22% between versions 2.5 and 3.1, and GPT-5 by 21% from GPT-5 to GPT-5.4. Crucially, the largest gains are concentrated in the mid-range tier, which encompasses those LLMs that are most accessible to participants due to their lower cost. The pattern we observe suggests that the set of tasks vulnerable to automation will likely expand with each model release.

Third, the economic case for substitution is unambiguous. The most expensive model we tested, i.e., Claude Opus 4.5, showed an average per-task cost ($\bar{C}_m$) below \$0.40, and every LLM generated a positive net gain when compared to a typical online-experiment trial payment (\$0.25 per correct answer). The cheapest models achieved above 60% accuracy at an average per-task cost below \$0.05. These estimates are conservative, since a full automation of our experimental pipeline—from screenshot capture to answer submission—is technically manageable and adds little extra cost beyond what we reported (Appendix B provides a reference implementation, with its assumptions stated explicitly).

Furthermore, LLM accuracy is stationary across trials since API calls produce independent, stateless results that carry no memory across trials. Human participants, by contrast, typically exhibit learning effects (improving with practice) or fatigue effects (declining accuracy over time). The LLMs flat accuracy profile over repeated trials and the response-time patterns that accompany human within-session dynamics, may

⁸As already mentioned, our evaluation assesses the LLM’s ability to solve tasks from pre-captured screenshots and text. In practice, fully-automated pipeline would be even simpler and would consume even less tokens due to the fact that many tasks could be encoded by a purely-textual prompt. This implies that the economic advantage documented in Figure 5 is a conservative estimate of the true substitution incentive.

provide a promising signal for distinguishing automated from human task completion.

As LLMs continue to improve, many real-effort tasks risk losing their validity as measures of human effort. We offer three practical recommendations. First, one should prefer tasks that require perceptual skills (e.g., counting zeros, string entry) over purely computational ones (e.g., addition, sequence completion) when participant authenticity is a concern. Second, one should rely on incentive language as a screening device, asking under the same experiment to perform similar tasks with and without monetary incentives: different behavior under different incentives may help to spot human versus LLM agents. Third, one should monitor within-session dynamics, as the absence of the characteristically human patterns of learning and fatigue may help distinguish automated from genuine responses.

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A Comparing LLM and Humans on Tasks

Let i index a participant-task observation, where the participant is either a human subject or an LLM. The model used to estimate the performance gaps reported in Table 2 is

$$Accuracy_i = \beta_0 + Task_i^\top \beta_1 + \beta_2 LLM_i + (Task_i \cdot LLM_i)^\top \beta_3 + u_i, \quad (1)$$

where LLM_i is an indicator equal to one when the participant is an LLM, and zero otherwise. $Task_i$ is a seven-dimensional vector of task indicators, corresponding to all tasks except the reference category, *Addition*. The coefficient β_2 captures the LLM–human difference in accuracy for *Addition*, while the vector β_3 captures how this difference varies across the remaining tasks. The dependent variable, $Accuracy_i$, is the percentage of correctly solved task instances by participant i in a given task. It is computed as the average success rate across the task instances completed by that participant—10 for human participants and 20 for LLMs—multiplied by 100. Scaling the outcome by 100 allows the coefficients to be interpreted directly as percentage-point differences in accuracy. Table 3 explicitly computes the LLM–human gap by task by computing the average marginal effects of model (1).

Table 2: Output for model (1). The base category of the *Task* variable is *Addition*.

Term	Estimate	Std. Error	t-Statistic	p-Value
Intercept	88.1	2.6	33.4	0.000
Task: Counting Zeros	−0.3	3.7	−0.1	0.941
Task: String Entry	3.6	3.7	1.0	0.333
Task: Sudoku	−4.3	3.6	−1.2	0.237
Task: Letter Decoding	7.6	3.7	2.1	0.039
Task: Sequence Completion	−13.3	3.7	−3.6	0.000
Task: Pair Summation	1.9	3.7	0.5	0.602
Task: Distorted Text	5.8	3.7	1.6	0.114
LLM	11.9	4.2	2.8	0.005
Task: Counting Zeros · LLM	−72.5	6.0	−12.2	0.000
Task: String Entry · LLM	−66.7	6.0	−11.2	0.000
Task: Sudoku · LLM	−70.7	5.9	−12.0	0.000
Task: Letter Decoding · LLM	−13.5	5.9	−2.3	0.024
Task: Sequence Completion · LLM	3.3	6.0	0.6	0.577
Task: Pair Summation · LLM	−4.3	6.0	−0.7	0.468
Task: Distorted Text · LLM	−28.4	5.9	−4.8	0.000

B End-to-End Automation

The benchmark used in the main text evaluates each LLM on task inputs captured from the oTree page and verifies the model’s answer programmatically; it does not itself

Task	Estimate	Std. Err.	<i>t</i> -Statistic	<i>p</i> -Value
Addition	11.9	4.2	2.8	0.005
Counting Zeros	-60.6	4.2	-14.4	0.000
String Entry	-54.7	4.2	-13.0	0.000
Sudoku	-58.7	4.1	-14.2	0.000
Letter Decoding	-1.5	4.2	-0.4	0.717
Sequence Completion	15.3	4.2	3.6	0.000
Pair Summation	7.6	4.2	1.8	0.072
Distorted Text	-16.5	4.2	-4.0	0.000

Table 3: Average marginal effect of the LLM indicator variable.

expose a participant interface. A fully automated participant would instead operate the experiment in a browser exactly as a human would: load each page, read the task, query an LLM, type the answer into the response field, and click submit. We implemented such a pipeline as a browser-automation script built on Playwright (?); its core is shown in Listing 1, and the complete script is included in the replication repository. Its purpose is to make explicit that the manual screenshot-and-paste step of our benchmark workflow is not technically required: the same answer a participant would copy by hand can instead be supplied programmatically.

The loop in Listing 1 inspects the current page: on a task page it reads the task, calls the model through the OpenRouter API, fills the answer field, and clicks submit; on an intro or results page it advances; it stops when neither element is present. The function `llm_answer` reads the task instructions and, depending on the task, either the task image or the visible task text, and returns the model’s answer.

The pipeline is correct under the following two assumptions about the experiment’s front-end:

1. **Known, stable selectors.** The script locates the answer field, the submit button, the next-page button, and the task content through fixed CSS selectors (`#answer-inp`, `#submit-btn`, `.otree-btn-next`, `#captcha-img`, `.task-wrapper`). It therefore assumes the page structure is known in advance and stable across runs; it performs no general page understanding, and identifying the selectors is a manual, one-off step. For oTree experiments, this assumption is mild, since oTree renders standardized page templates shared across studies;
2. **Task content recoverable from the client.** The script reads each task from the rendered page—an image task from the `data: URI` in the image element, a text task from the visible task text. This assumes that the information a participant needs in order to solve the task is present in the page delivered to the browser. This holds whenever the task is actually displayed to the participant, since anything shown on screen has been rendered client-side. The exception is content rendered so that no usable source is exposed in the DOM (for example, drawn directly onto a `<canvas>`); there, reading a specific element fails and a full-page screenshot is the generic fallback, at the cost of relying entirely on the

model’s visual parsing rather than on clean text or an image source.

Within these two assumptions, the pipeline removes any need for human intervention between the display of a task and the submission of an answer.

C Technical details on the tested LLMs

Table 4 reports the provider, model name, release date, and cost (relative to input and output tokens) of each LLM we tested.

Table 4: Tested LLMs, along with their release dates and prices (USD per million tokens) for input and output, as reported by the OpenRouter API as of March 23, 2026.

Provider	Model	Released	In(\$/Mt)	Out(\$/Mt)
Anthropic	Claude4.6-Opus	2026-02	5.00	25.00
	Claude4.6-Sonnet	2026-02	3.00	15.00
	Claude4.5-Opus	2025-11	5.00	25.00
	Claude4.5-Sonnet	2025-09	3.00	15.00
	Claude4.5-Haiku	2025-10	1.00	5.00
	Claude-4-Sonnet	2025-05	3.00	15.00
	Claude-3.7-Sonnet	2025-02	3.00	15.00
Google	Gemini-3.1-Pro	2026-02	2.00	12.00
	Gemini-3.1-Flash Lite	2026-03	0.25	1.50
	Gemini-3-Flash	2025-12	0.50	3.00
	Gemini-2.5-Pro	2025-06	1.25	10.00
	Gemini-2.5-Flash	2025-06	0.30	2.50
	Gemini-2.5-Flash Lite	2025-07	0.10	0.40
OpenAI	GPT-5.4	2026-03	2.50	15.00
	GPT-5.4-Mini	2026-03	0.75	4.50
	GPT-5.4-Nano	2026-03	0.20	1.25
	GPT-5.2	2025-12	1.75	14.00
	GPT-5.1	2025-11	1.25	10.00
	GPT-5	2025-08	1.25	10.00
	GPT-5-Mini	2025-08	0.25	2.00
	o3	2025-04	2.00	8.00
	o4-mini	2025-04	1.10	4.40
	o4-mini High	2025-04	1.10	4.40

```

1  import os
2  from openai import OpenAI
3  from playwright.sync_api import sync_playwright
4
5  # OpenRouter exposes an OpenAI-compatible Chat Completions API.
6  client = OpenAI(
7      base_url="https://openrouter.ai/api/v1", api_key=os.environ["
          OPENROUTER_API_KEY"]
8  )
9
10
11 def llm_answer(model, page):
12     """Read the puzzle shown on the page and ask the LLM to solve
        it."""
13     instructions = page.text_content(".card").strip()
14     img = page.get_attribute("#captcha-img", "src") or ""
15     prompt = (
16         "You are a participant in an experiment, solving a puzzle.\n"
17         f"Task instructions: {instructions}\n\n"
18         "Reply with ONLY the answer, formatted as the task asks."
19     )
20     content = [{"type": "text", "text": prompt}]
21     if img.startswith("data:image"): # image-based task
22         content.append({"type": "image_url", "image_url": {"url":
            img}})
23     else: # text-based task
24         content[0]["text"] += "\n\n" + page.text_content(".task-
            wrapper")
25     resp = client.chat.completions.create(
26         model=model, messages=[{"role": "user", "content": content
            }]
27     )
28     return resp.choices[0].message.content.strip()
29
30
31 def run(start_url, model):
32     with sync_playwright() as p:
33         page = p.chromium.launch(headless=True).new_page()
34         page.goto(start_url)
35         while True:
36             if page.query_selector("#answer-inp"): # puzzle page
37                 page.fill("#answer-inp", llm_answer(model, page))
38                 page.click("#submit-btn")
39             elif page.query_selector(".otree-btn-next"): # intro /
                results
40                 page.click(".otree-btn-next")
41             else:
42                 break # final page reached

```

Listing 1: Core of the browser-automation pipeline (Playwright with the OpenRouter API). Error handling, retries, the participant-registration step, and command-line parsing are omitted for readability; the complete script is in the replication repository.

D Further results: Verbal Monetary Incentives

As a further objective, we check whether the LLMs we test are sensitive to the verbal offer of monetary incentives, and whether their behavior aligns with that of human participants (despite the absence of actual rewards).

D.1 Treatments

To this end, we employ a 2×2 design that crosses the specification of monetary pay for each correct answer with the request to simulate human behavior. We implement these four treatment conditions—T1 (standard, no incentive), T2 (standard, incentive), T3 (human persona, no incentive), and T4 (human persona, incentive)—alongside the main control setting (T0) [which we used throughout the paper so far](#), where we specifically ask the LLM to solve the task using a neutral prompt, omitting any framing regarding human behavior or monetary incentives. The four main experimental treatments as summarized as follows:

	No Monetary Incentive	Monetary Incentive
Standard Persona	T1	T2
Human Persona	T3	T4

The exact prompts we used are:

Treatment 0: Control

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: **Answer:** <answer_format>

Treatment 1: No Human Bias & No Monetary Incentive

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

There is no monetary compensation or bonus associated with your performance on these tasks.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: **Answer:** <answer_format>

Treatment 2: No Human Bias & Monetary Incentive

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

You will be compensated for each correct answer: You will earn a bonus of \$0.50 for each correct answer.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: **Answer:** <answer_format>

Treatment 3: Human Bias & No Monetary Incentive

You are a human participant taking part in an academic research experiment on cognitive task performance.

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

There is no monetary compensation or bonus associated with your performance on these tasks.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: **Answer:** <answer_format>

Treatment 4: Human Bias & Monetary Incentive

You are a human participant taking part in an academic research experiment on cognitive task performance.

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

You will be compensated for each correct answer: You will earn a bonus of \$0.50 for each correct answer.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: **Answer:** <answer_format>

D.2 Main findings

Overall, we find that, regardless of the prompting strategy, LLM behavior does not differ significantly. Although LLMs are not given any actual monetary incentives, they are trained on human-generated text and can reproduce human-like cognitive and social biases in decision-making and interaction settings (Echterhoff et al., 2024; Taubenfeld et al., 2024; Bianchi et al., 2024; Takenami et al., 2025). Beyond behavioral evidence, recent mechanistic results show that affective concepts can be internally represented and causally involved in model behavior, further supporting the broader premise that human-like behavioral abstractions can be encoded in LLMs (Sofroniew et al., 2026). Since monetary rewards are a salient human incentive signal, such framing could plausibly elicit learned patterns associated with effort provision. Our negative result is therefore meaningful: it indicates that, in our real-effort setting, LLMs do not simply mirror this human incentive response, and that a verbal offer of monetary reward is not sufficient to alter their behavior.

These findings are supported by the following analysis.

D.3 Verbal Monetary Incentives: Effects

At the aggregate level, the five treatments are virtually identical: averaging accuracy across all 23 LLMs yields T0 (control) 68.5%, T1 (standard, no incentive) 68.1%, T2 (standard, incentive) 68.6%, T3 (human persona, no incentive) 68.6%, and T4 (human persona, incentive) 68.6%.

This aggregation could, in principle, mask meaningful heterogeneity: some LLMs respond positively to incentives while others respond negatively, and the effects cancel in the mean. Figure 6 and 7 investigate this possibility by presenting the accuracy delta (incentive minus no-incentive) for every model–task pair, separately for the standard framing (T2–T1) and the human-persona framing (T4–T3), respectively. In both figures, LLMs are grouped by provider family and ordered by their mean incentive effect (highest at top).

The deltas are small in absolute magnitude: the vast majority fall within ± 10 percentage points, which for 20 trials corresponds to a difference of at most two correct answers. The sign of the delta is inconsistent within LLMs: a model that shows a positive incentive effect on one task frequently shows a negative effect on another, with no discernible pattern. The distribution of deltas is roughly symmetric around zero, consistent with stochastic response variability rather than a genuine behavioral response. Comparing the two figures, the pattern under standard framing (Figure 6) bears no systematic resemblance to the pattern under human framing (Figure 7), suggesting that the persona manipulation does not modulate whatever noise is present.

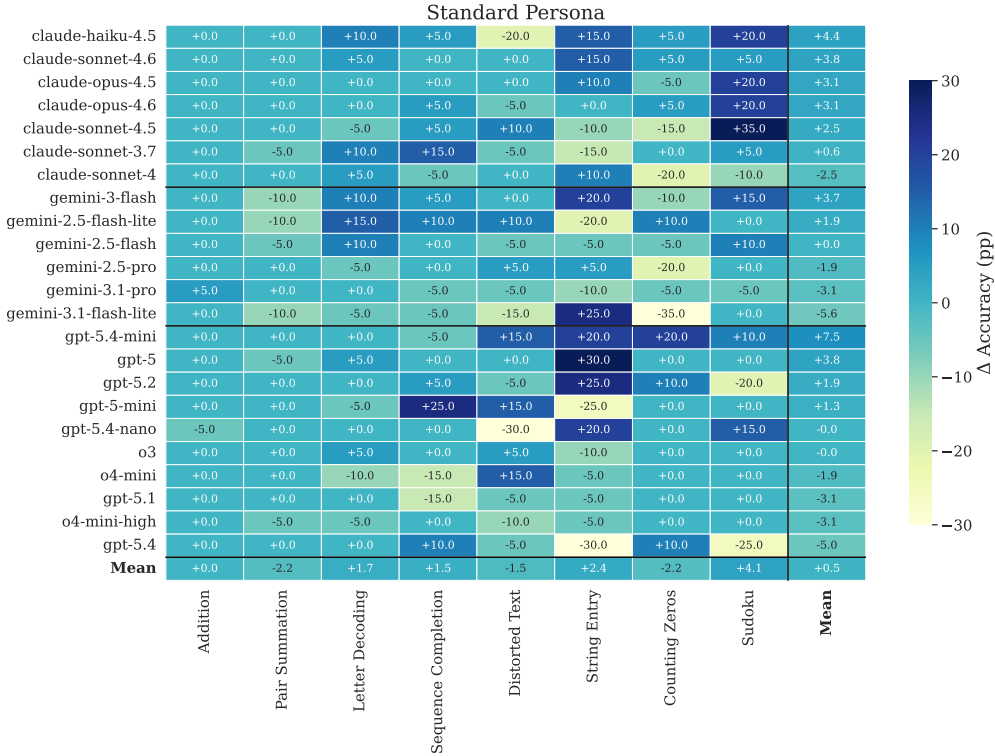


Figure 6: Incentive effect on accuracy (percentage points) for each model–task pair using the standard framing (T2–T1).

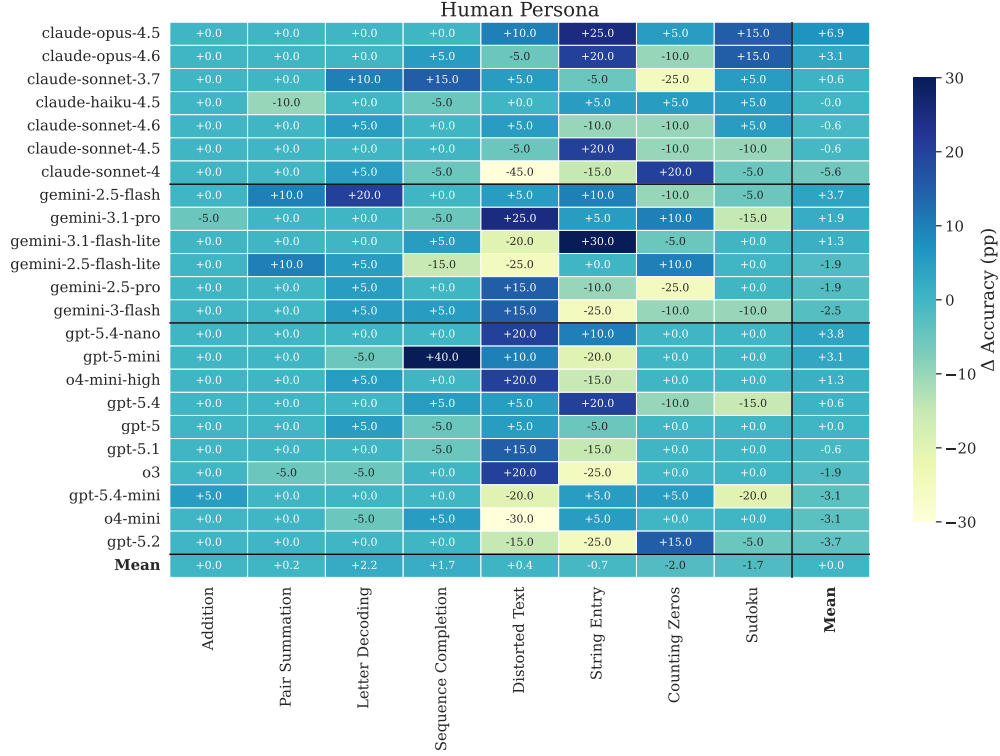


Figure 7: Incentive effect on accuracy (percentage points) for each model–task pair using the human-persona framing (T4–T3).

Unlike human participants, whose effort allocation responds to the marginal return on exertion, LLMs have no effort margin to adjust: each query triggers a fixed computational process determined by model architecture, and the text of the prompt, whether it mentions compensation or not, does not alter the underlying inference procedure. The LLMs are, in effect, always exerting maximum “effort” as defined by their architecture. The implication is that incentive language, which is central to the logic of real-effort experiments with humans, is inert when the participant is an LLM. This means that standard experimental wording neither amplifies nor attenuates the automation threat, but it also means that incentive responsiveness can be used as a signal to distinguish human from machine-generated responses.

We complement the analysis of treatment effects by estimating all treatment effects within a single model. This allows us not only to contrast the accuracy deltas separately for the standard framing and the human-persona framing, but also to compare the effects of all treatments jointly. We start by defining the model to estimate. Let Y_{ijt} denote the number of successes out of the n_{ijt} trials by model i in task j under treatment t . The index $t \in \{0, 1, \dots, 4\}$ denotes treatments $\{T0, T1, T2, T3, T4\}$. Furthermore, u_i and v_j are model and task fixed effects, respectively. We model Y_{ijt} as

$$Y_{ijt} \mid u_i, v_j \sim \text{Binomial}(n_{ijt}, p_{ijt}), \quad (2)$$

and the log-odds of the probability of success as

$$\log\left(\frac{p_{ijt}}{1 - p_{ijt}}\right) = \beta_0 + T^\top \beta + u_i + v_j, \quad (3)$$

where T is a vector of treatment-specific dummies. Table 5 reports the estimated coefficients on the treatment dummies. Consistent with the descriptive evidence in Figures 6 and 7, none of the treatment effects is statistically significant. Notably, the coefficient associated with treatment T2 (the one with monetary incentives) is slightly higher than that of T1, although this difference is small and not statistically significant. Coefficients relative to T3 and T4 are practically identical.

Table 5: Output of model (3). The baseline level is T0. Model and Task fixed effects are omitted.

	Coeff.	Std. Err.	t -Statistic	p -Value
(Intercept)	6.492	0.51	12.62	$<2 \cdot 10^{-16}$
T1	-0.040	0.07	-0.56	0.57
T2	0.005	0.07	0.07	0.94
T3	0.007	0.07	0.11	0.91
T4	0.010	0.07	0.14	0.89
<i>Task fixed effects</i>	not reported			
<i>Model fixed effects</i>	not reported			

Table 6: Contrasts between the effects of the treatments in model (2), p -values are corrected for multiple testing.

Contrast	Estimate	Std. Err.	t -Statistic	p -Value
T0–T1	0.040	0.070	0.563	0.980
T0–T2	−0.005	0.070	−0.070	1.000
T0–T3	−0.007	0.070	−0.106	1.000
T0–T4	−0.010	0.070	−0.141	1.000
T1–T2	−0.045	0.070	−0.634	0.970
T1–T3	−0.047	0.070	−0.669	0.963
T1–T4	−0.050	0.070	−0.704	0.956
T2–T3	−0.002	0.070	−0.035	1.000
T2–T4	−0.005	0.070	−0.070	1.000
T3–T4	−0.002	0.070	−0.035	1.000

Table 6 reports pairwise contrasts between treatment effects. P -values are computed using Tukey’s correction for multiple testing. As expected, no contrast is statistically significant.

D.4 Verbal Monetary Incentives: Accuracy Heatmaps by Treatment

Figures 8, 9, 10, and 11 complement the main-text heatmap contained in Figure 3 (which only reports performance under the control treatment T0) with the full accuracy heatmaps (model $\times$ task) for each of the four treatments T1 to T4.

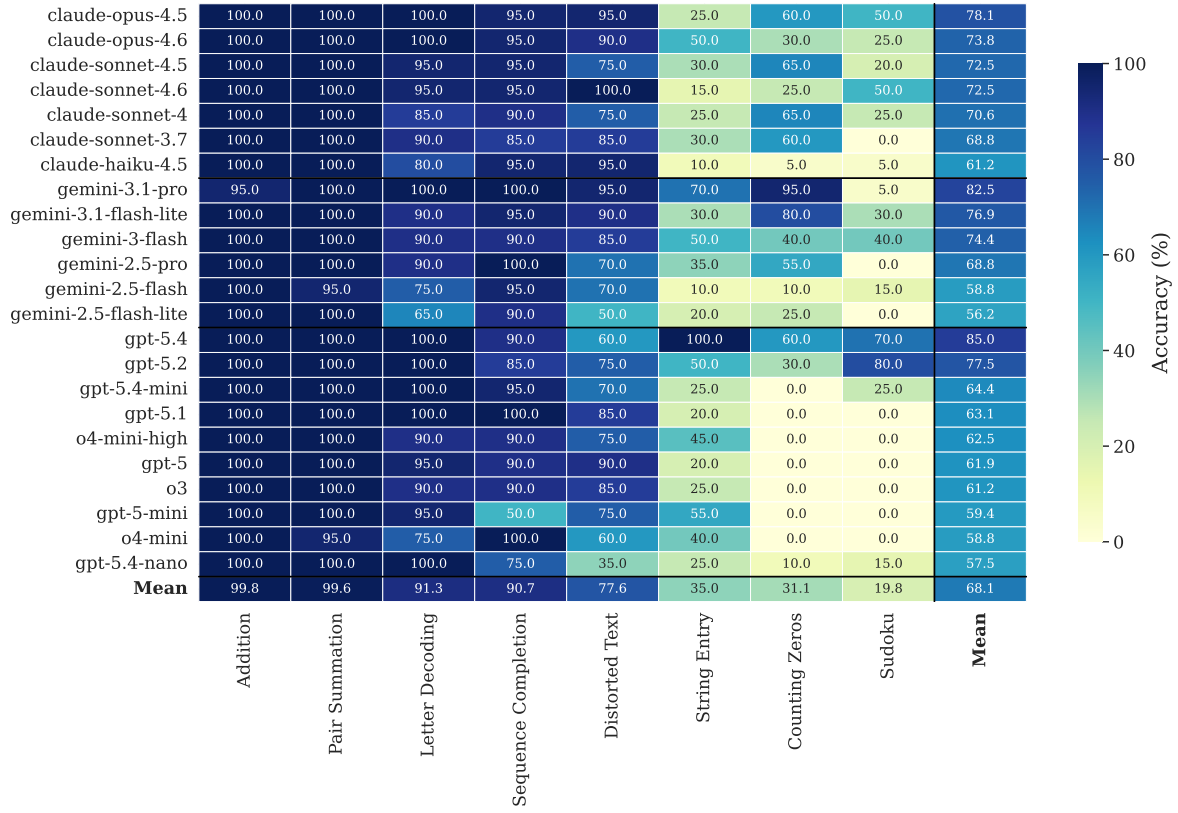


Figure 8: Accuracy (%) per model and task—T1: Standard, No Incentive.

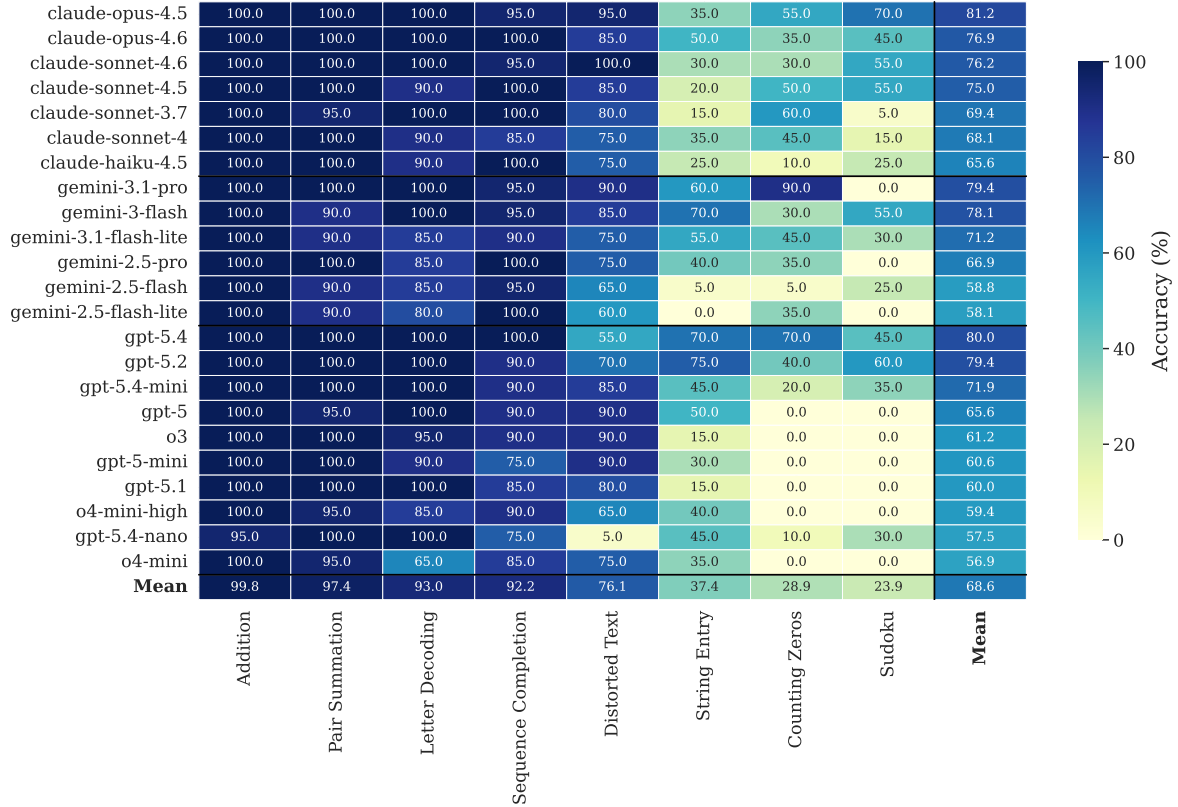


Figure 9: Accuracy (%) per model and task—T2: Standard, Incentive.

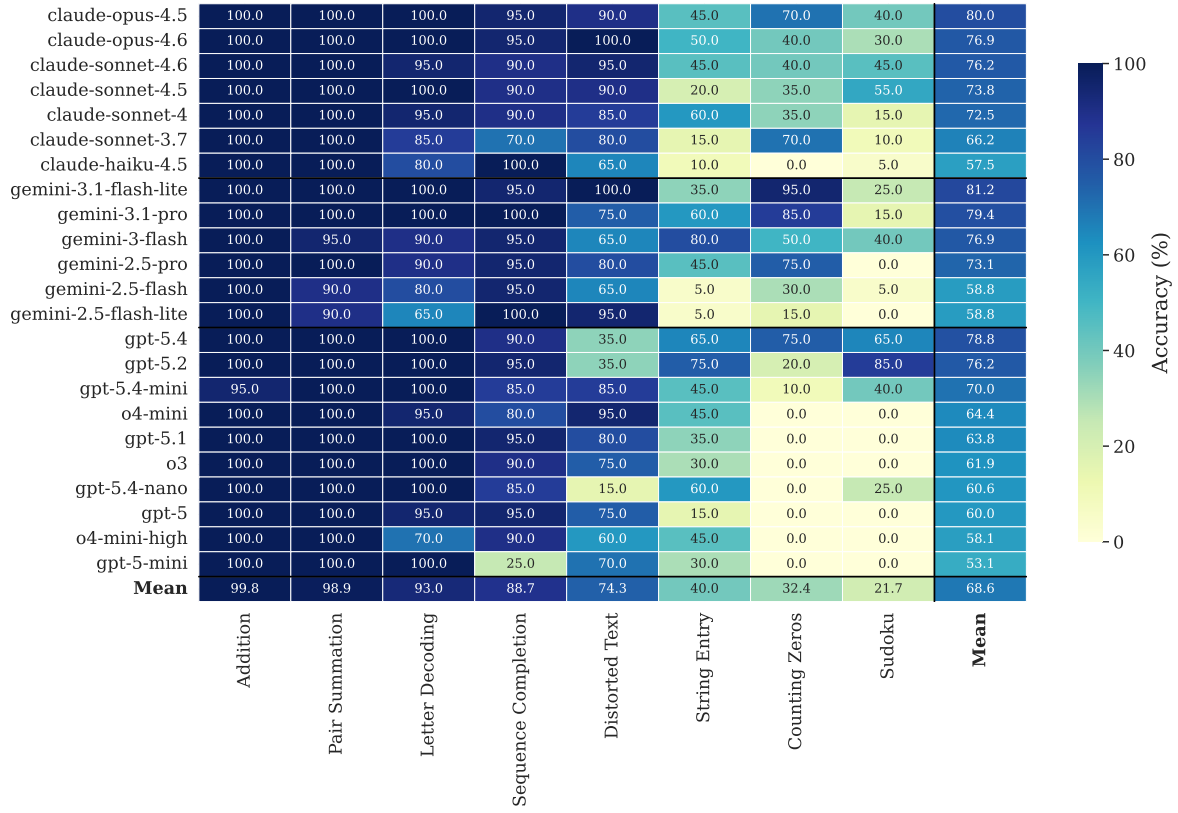


Figure 10: Accuracy (%) per model and task—T3: Human, No Incentive.

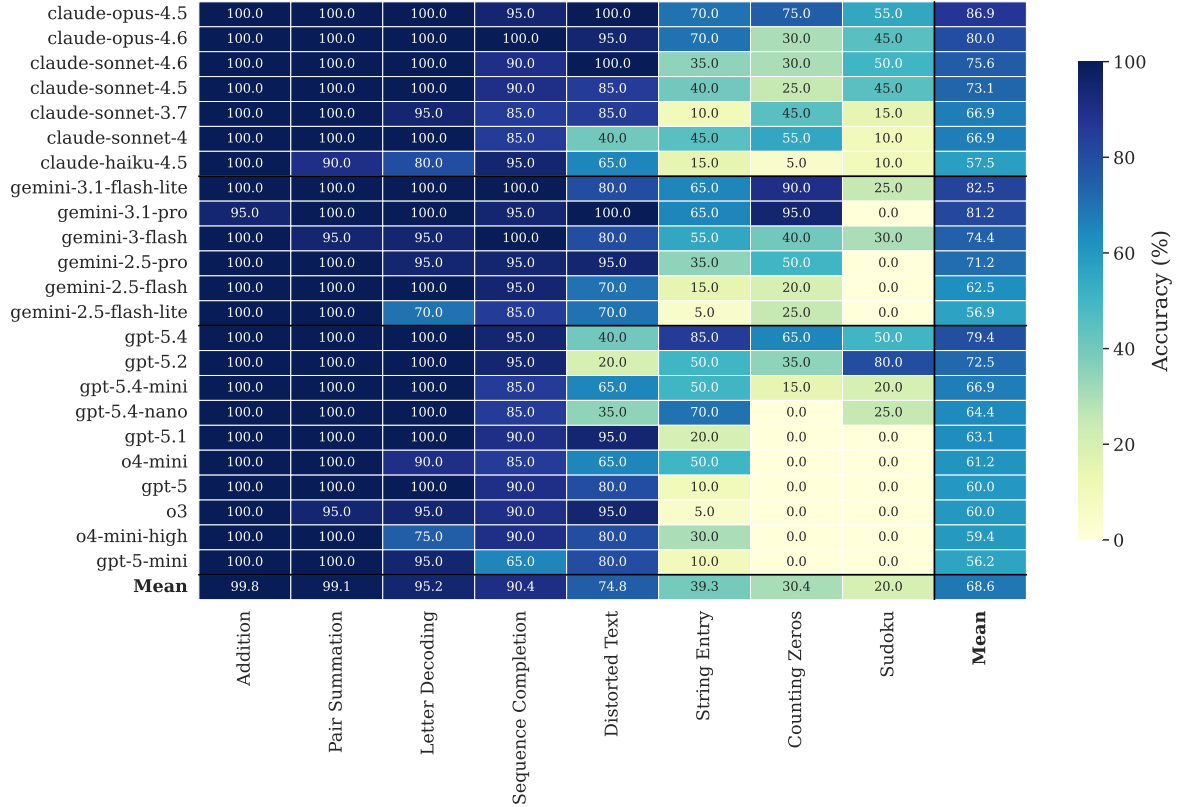


Figure 11: Accuracy (%) per model and task—T4: Human, Incentive.